23. Number Sequences & Patterns

Year 4 · **Supplementary Worksheet** — extra practice & worked breakdowns

Worked Examples

Example 1 — Finding the next terms:

Find the next two terms: 5, 10, 15, 20, __, __.

Answer: The rule is add 5 each time. Next terms: **25, 30**.

Example 2 — Finding a missing term:

Find the missing term: 6, 12, __, 24, 30.

Answer: The rule is add 6 each time, so the missing term is 12 + 6 = **18**.

Example 3 — A growing pattern:

A pattern of counters has 4 counters in step 1, 8 in step 2, 12 in step 3. How many counters are in step 5?

Answer: The pattern goes up by 4 each step: 4, 8, 12, 16, 20. Step 5 has **20 counters**.

Example 4 — Doubling sequence:

Find the next term: 2, 4, 8, 16, __.

Answer: Each term doubles the one before: 16 × 2 = **32**.

Example 5 — Multi-step pattern word problem:

One table seats 5 people. Each extra table pushed on adds 3 more seats. How many people can a row of 4 tables seat?

Answer: 1 table seats 5. Each extra table (3 more) adds 3: table 2 → 8, table 3 → 11, table 4 → **14 people**.

Foundation

15 questions

1. Find the next two terms: 4, 8, 12, 16, __, __.

2. Find the next two terms: 20, 30, 40, 50, __, __.

3. Which number continues the sequence 3, 6, 9, 12, __? A) 13 B) 14 C) 15 D) 16

4. Find the missing term: 5, 10, __, 20, 25.

5. Find the next two terms: 60, 55, 50, __, __.

6. Find the next two terms: 2, 4, 6, 8, __, __.

7. Select ALL the sequences below that count up in steps of 3: A) 3, 6, 9, 12 B) 2, 4, 6, 8 C) 9, 12, 15, 18 D) 5, 10, 15, 20 E) 21, 24, 27, 30

8. Find the missing term: 3, 6, __, 12, 15.

9. Find the next two terms: 90, 80, 70, __, __.

10. Which of these is the rule for the sequence 7, 14, 21, 28...? A) add 5 B) add 7 C) add 9 D) multiply by 2

11. Find the next two terms: 6, 9, 12, 15, __, __.

12. Find the missing term: 40, 35, __, 25, 20.

13. Select ALL the numbers that could continue the sequence 8, 16, 24, 32…: A) 40 B) 35 C) 48 D) 45 E) 56

14. Find the next two terms: 6, 12, 18, __, __.

15. Which number is missing? 10, 14, 18, __, 26. A) 20 B) 21 C) 22 D) 23

Standard

20 questions

1. Find the next two terms: 3, 6, 12, 24, __, __.

2. Find the next two terms: 2, 3, 5, 8, __, __. (Hint: look at how the gap between each term changes.)

3. What is the rule for the sequence 4, 8, 16, 32…? A) add 4 B) add 8 C) double each time D) add 16

4. Find the missing term: 3, 7, __, 15, 19.

5. Find the next two terms: 160, 80, 40, __, __.

6. A pattern of squares has 2 squares in row 1, 4 squares in row 2, 6 squares in row 3. How many squares are in row 4?

7. Select ALL the sequences below that step up by 5 each time: A) 5, 10, 15, 20 B) 4, 8, 12, 16 C) 10, 15, 20, 25 D) 6, 10, 14, 18 E) 20, 25, 30, 35

8. Find the missing term: 11, 14, __, 20, 23.

9. Find the next two terms: 85, 75, 65, __, __.

10. Which of these is the missing term? 50, 43, 36, __, 22. A) 27 B) 28 C) 29 D) 30

11. Find the next two terms: 12, 24, 36, 48, __, __.

12. A pattern of dots doubles each time: 5, 10, 20, 40, __. What is the next term?

13. Find the missing term: 90, 75, __, 45, 30.

14. Which rule describes the sequence 2, 5, 10, 17…? A) add consecutive odd numbers B) add 3 each time C) multiply by 2 each time D) add 4 each time

15. Select ALL the numbers below that are square numbers: A) 4 B) 10 C) 16 D) 20 E) 36

16. Find the next two terms: 1, 2, 4, 8, __, __.

17. Find the missing term: 6, 13, __, 27, 34.

18. Find the next two terms: 75, 65, 55, __, __.

19. Which rule describes the sequence 100, 91, 82, 73…? A) subtract 7 B) subtract 9 C) subtract 11 D) subtract 5

20. Select ALL the numbers below that are square numbers: A) 9 B) 14 C) 25 D) 30 E) 49

Hard

5 questions

1. A pattern of triangles is made from matchsticks: 1 triangle uses 3 sticks, and each extra triangle added in a row needs 2 more sticks. How many sticks for 5 triangles?

2. Find the next two terms: 1, 4, 9, 16, 25, __, __.

3. Select the TWO sequences below that double each time: A) 3, 6, 12, 24 B) 2, 4, 6, 8 C) 5, 10, 20, 40 D) 7, 14, 21, 28 E) 4, 8, 12, 16

4. A sequence starts at 60 and decreases by 7 each time. What is the 5th term?

5. A garden path is made from square slabs: 1 slab needs 4 edges, and each extra slab added in a row needs 3 more edges. How many edges for 6 slabs?

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